

PETadex: A comprehensive database of plastic-degrading enzyme sequence diversity

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Abstract

Introduction

Since plastics started being produced in the mid-20th century, they have proven invaluable in improving the accessibility and affordability of countless goods such as medical supplies and food packaging. However, their low production costs and durability have also made plastics a persistent large-scale pollutant [plastic pollution citation here]. [X]% of the estimated 13 billion tonnes of globally produced plastic waste are managed through recycling or incineration, but the fate of the rest remains unaccounted for [plastic pollution citation here] (Fig. 1A). Yet, plastic continues to be mass-produced at an alarming rate. Already, there is more than 2.6 times more carbon trapped in plastics than in all living animals on Earth (Fig. 1C): human plastic production is having a geological impact on the planet.

Despite the relative novelty of plastics on Earth, scientists have discovered several organisms which have adapted to break down plastics using plastic-degrading enzymes (PDEs). In 2016, Yoshida *et al.* discovered *IsPETase*, an efficient PDE capable of breaking down the common plastic Polyethylene Terephthalate (PET) into its constituent monomers, which are small molecules that make up PET. (Fig. 1B–D). These monomers can then be used for repolymerization of new

plastic products, or disposed of via biodegradation. This discovery sparked a surge of research in PDEs, which represent a possible solution to the problem of plastic waste.

Convolutional Neural Networks (CNNs) trained on generic non-PDE datasets were used to predict 5 thermo-stabilizing mutations of *IsPETase*, called *FAST-PETase*, which improved activity 38-fold at 50°C [cite]. Sequence-based mining for the discovery of new enzymes is also well established method, with Danso *et al.*, 2018 [cite] having discovered 504 candidates from database searches on UniprotKB and NCBI-nr, alongside 349 candidates from mining 133 metagenomes. More recent attempts, such as Erickson *et al.*, 2022 [cite] expanded by screening 87 targeted hot-spring metagenomes (~38 million sequences) for plastic-degrading enzymes, characterizing 74 putative enzymes, and further testing 51 for activity, and characterizing 23 novel plastic-active enzymes.

However, the diversity of plastic-degrading enzymes is likely much larger than what has been discovered so far. A notable database, the plastics-active enzymes database (PAZy), contains 495 functionally verified plastic-active enzymes. Public sequence databases like the Sequence Read Archive (SRA) contains over 27 million globally diverse samples, and are largely an untapped

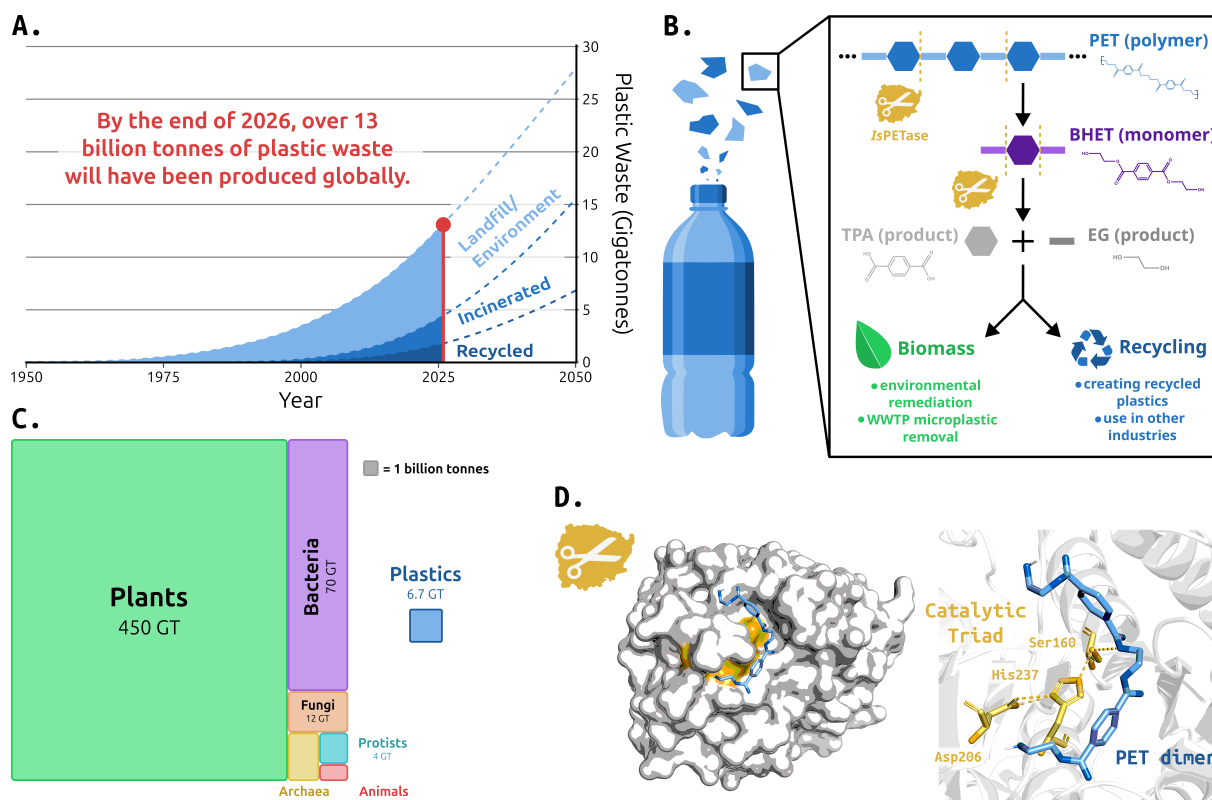


Fig. 1 Plastic waste and its enzymatic degradation. (A) Cumulative global plastic waste by fate, 1950–2050. (B) IsPETase hydrolyses PET to the monomer BHET, then to TPA and EG, feeding biomass or recycling. (C) Carbon mass by category; plastics (6.7 Gt) exceed all animals (2.5 Gt) 2.6-fold. (D) IsPETase active site: Ser160–His237–Asp206 triad with bound PET dimer.

resource for the discovery of novel plastic-degrading enzymes. We recently built Logan [cite], a tool that assembled the entire SRA and retrieved 215.7M putative plastic-degrading enzymes. We demonstrated that the discovery of novel plastic-degrading enzymes within this dataset is possible through operationally random selection. However, the lack of annotations prevents strategic selection of putative plastic-degrading enzymes for validation.

Here we present PETadex, a methodological and quantitative rebuild of our previous dataset. This novel dataset supports the targeted selection of enzymes with plastic-degrading activity. Using a carefully curated input dataset of 411 plastic-degrading enzymes, we searched NCBI-nr and the SRA (using the Logan assembly), constraining results using catalytic-residue-constrained profile HMMs. We built a dataset of [x] million

enzymes that carry the catalytic domain boundaries, annotated with fold-assignments, protein-structures and the temperature/pH of the biosamples of origin. This enables candidate plastic-degrading enzymes to be filtered based on catalytic integrity, structural fold, and metadata properties prior to testing. This enables the high throughput targeted-selection of candidates, a process which can be further automated using innovative machine learning methods. The dataset of enzymes, annotations and a toolkit of applicable methods are freely available at <https://petadex.net> for the purpose of catalyzing discovery efforts worldwide.

Results

Sequence and structure-based annotation of 609 known plastic-degrading enzymes

Following established workflows for the homology-based retrieval and expansion of a given set of enzymes of interest [cite Erickson, Danso, Logan etc.], we built a workflow for the retrieval of plastic-degrading enzymes from NCBI-nr and the SRA (using the Logan assembly).

Improving upon our previous PETadex dataset, by updating the input search query and fixing several methodological limitations, we present the updated PETadex v2 dataset.

To generate this updated PETadex-260701 dataset, we created a starting search query from combining two versions of the PAZy database, the first from the original PETadex generation (pazy-240826; $n = 212$; August 26th, 2024), and a recently updated version (pazy-260701; $n = 473$; July 1st, 2026). Sourcing 136 enzymes from pazy-240826, 397 enzymes from pazy-260701 and 76 merged between both, generating a 609 enzyme union set (plasticome). Plasticome was then clustered at 90% amino acid identity to 411 cluster centroids (Fig. 2A). Predicting structures of these centroids using ESMFold2 (ESMC-6B trunk), then searches against the CATH S40 database with Foldseek alongside searching against InterProScan resolved the proteins into 15 distinct protein domains. These centroids are further aligned into 46 graph components ($\geq 30\%$ amino acid identity and $E < 1 \times 10^{-5}$) (Fig. 2A).

Homology-based retrieval yields a dataset of 215.7 million putative plastic-degrading enzymes

To expand the dataset of known plastic-degrading enzymes beyond the plasticome set, we first searched these 411 centroids into the NCBI nr database using DIAMOND(DIAMOND v2.2.4.184, `--very-sensitive`, `-k0`), recovering 42,053,081 pairwise alignments, collapsing to 8,850,583 unique sequences [cite]. Of these, 7,825,535 pass the e-value filter ($E \leq 1 \times 10^{-5}$), and further removing plasticome self-hits returns a final count of 7,825,107 sequences (Fig. 2B).

In order to search Logan with only the catalytic domains of these plasticome homologs, we built profile HMMs to characterize the plastic-degrading domain boundaries and catalytic residues. These HMMs were built from seed MSAs (MUSCLE 5.3) of plasticome sequences together with their closest nr homologs ($E \leq 1 \times 10^{-50}$), where each member of the alignments must contain the residues involved in the catalytic mechanism, annotated from literature [cite]. The 46 graph components were further segmented using these catalytic mechanisms, resolving into 53 MSAs. The HMMs were built from the seed MSAs using a pipeline analogous to JACKhmmer [cite]. HMMs were created using HMMbuild, and then searched (HMMsearch) against close NCBI-nr homologs ($E \leq 1 \times 10^{-10}$), annotating the catalytic residues [cite]. Alignments were created from sequences that were below the e-value cut-off of 1.0E-10 and contained all of the canonical catalytic residues. These MSAs serve as the seed MSAs for the next iteration of the pipeline. While creating the next iteration’s HMMs, match and insert state annotations were propagated from the previous HMM and any column of the alignment that had an occupancy below 0.7 was declared an insert state. The pipeline was then repeated for a total of 4 iterations.

Searching these component HMMs against the 7,825,107 nr sequences yielded 7,562,397 HMM hits. Of these hits, the HMM range of 6,857,582 contained all of the positions of the catalytic residues ("complete hits"), of which 6,343,582 contained the canonical catalytic residue identities. The ranges annotated as the catalytic domains were then clustered (DIAMOND `-linclust`), resulting in 2,740,458 90% centroids (Fig. 2B). These centroids serve as the input query into Logan.

Expanding on the nr search, we queried these [x] million enzyme homologs across Logan (version [x]) contigs, recovering [x] billion matching sequences (Fig. 2B). Of these Logan sequences, [x] matched an nr enzyme at 90% identity, and clustering the remaining [x] at 90% aaid, yielding [x] million non-redundant enzyme homologs (Fig. 2C,D). We’ve dubbed this dataset PETadex-v2, with sub sets dubbed PETadex-plasticome (or simply plasticome), PETadex-nr and PETadex-logan, depending on the origin. [Describe a

Fig 2A. Plasticome Composition



Fig 2B. PETadex Generation

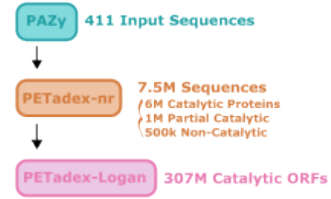


Fig 2C. PETadex Clustering

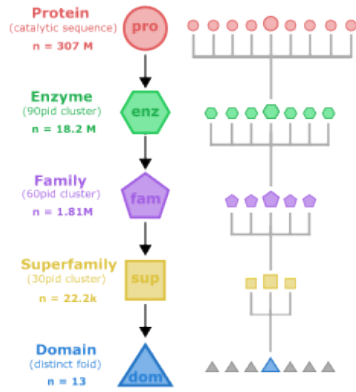


Fig 2D. Cluster Size Distribution

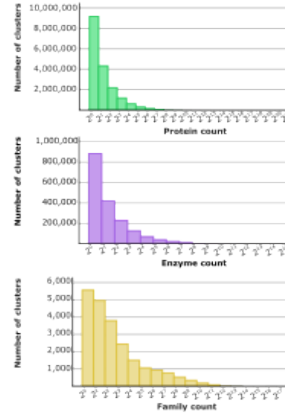


Fig. 2 The PETadex-260701 dataset. (A) Composition of the plasticome: the 609-enzyme union of the two PAZy snapshots, its collapse to 411 cluster centroids at 90% amino acid identity, the 46 graph components those centroids resolve into, and the CATH folds assigned to them, coloured by provenance. (B) Overview of PETadex generation, from the 411 input sequences through the PETadex-nr and PETadex-logan expansions. (C) Hierarchical clustering of catalytic sequences into enzymes (90% identity), families (60%), superfamilies (30%) and distinct folds. (D) Cluster size distributions at the protein, enzyme and family levels.

domain from the search, can be a/B hydrolase, Amydases, etc.]

Structural, environmental and taxonomic annotations build the PETadex enzyme space

To explore the PETadex dataset, we annotated its sequences along three axes: structural, environmental and taxonomic.

We first predicted structures for the 22,235 superfamily centroids (30% amino acid identity)

using ESMFold2 (ESMC-6B trunk). These structures resolve both the catalytic domain and the accessory domains flanking it, which may additionally contribute to the catalytic mechanism.

We next enriched the dataset with environmental metadata from the record each sequence was assembled from, recovering geographical origin for [x] sequences (Fig. 3B) alongside conditions documented at collection, including temperature ([x] biosamples) and pH (9,287 biosamples). This distribution reflects where sequencing has been performed as much as where these enzymes occur.

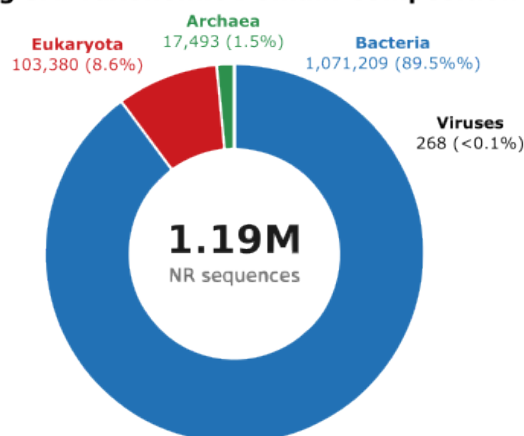
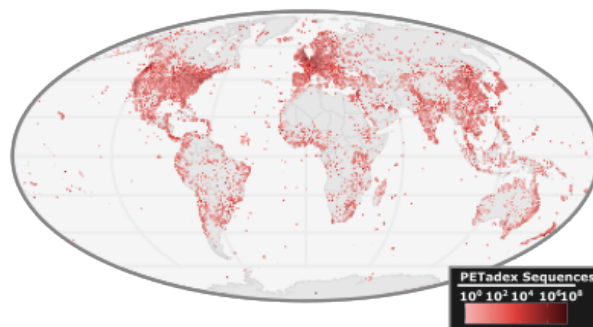
Fig 3A. Taxonomic Domain Composition**Fig 3B. Global Plastic-degrading Enzyme Distribution**

Fig. 3 Metadata annotations across the PETadex dataset. (A) Taxonomic domain composition of the 1.19 million taxonomically annotated nr sequences: Bacteria (89.5%), Eukaryota (8.6%), Archaea (1.5%) and Viruses (<0.1%). **(B)** Global geographic distribution of PETadex sequences by biosample collection site, coloured on a log scale.

Finally, we assigned each sequence a source organism, from the NCBI taxonomy of its GenBank record (PETadex-nr) or the taxonomic composition of its source run (PETadex-logan). Of [x] PETadex-nr sequences, 1,192,350 ([x]%) carry an assignment, overwhelmingly bacterial (1,071,209; 89.5%), followed by Eukaryota (103,380; 8.6%), Archaea (17,493; 1.5%) and Viruses (268; <0.1%) (Fig. 3A). Joining these taxa to cultivated growth optima further yields an inferred growth temperature and pH for [x] and 3,463,106 biosamples respectively [cite BacDive]. As averages over cultivated organisms, these are compressed toward mesophilic, neutral values ([T_{env} range]), and we treat them as distinct from the conditions recorded at collection.

Together these annotations support homology-based characterization of uncharacterized enzymes, and allow candidates to be filtered on catalytic integrity, fold and habitat prior to testing.

Website

This homology-based characterization pipeline is implemented through a DIAMOND sequence search interface available freely at petadex.net. This sequence search runs 20 shards in parallel

to achieve an average search time of ~47 seconds over the database of ~215.7 million enzymes.

Discussion

Methods

Supplementary information.

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Declarations

- Funding
- Competing interests
- Ethics approval and consent to participate
- Consent for publication
- Data availability
- Materials availability
- Code availability
- Author contribution

References